

DEXTER FERNANDES

Senior Computer Vision Engineer Bristol, UK +44 7436 007369 dexterfernandes11@gmail.com linkedin.com/in/dexterferns
github.com/Dexter-Fernandes

SUMMARY

Computer Vision Engineer with 3+ years building and deploying production video analytics, object detection and multi-object tracking systems. Led engineers and owned end-to-end CV pipelines across data curation, model development, validation, edge deployment and monitoring. Built multi-camera systems with PyTorch, OpenCV, NVIDIA DeepStream and OpenVINO, delivering measurable gains in accuracy, throughput, reliability and labelling automation.

RELEVANT EXPERIENCE

Esbaar | AI

01/2023 - 08/2023 | MUSCAT, OMAN

Engineer

- Owned the ML lifecycle for real-time perception on BP drilling rigs, from data curation and detector training through edge deployment, on-site validation and drift monitoring, improving F1 by 15% under 24/7 operation.
- Developed and maintained NVIDIA DeepStream 6.x and FFmpeg video pipelines covering camera ingestion, inference and field diagnostics across live rig deployments in desert conditions.
- Designed controlled training experiments for occlusion, lighting variation and motion artefacts, then measured F1 and failure rate to improve robustness within three months.
- Built ingestion and validation pipelines across image, video and sensor data with Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout time from 72 hours to under 24 hours.
- Benchmarked YOLOv5 against EfficientDet on COCO-format production data, measuring accuracy, latency and stability before and after optimisation; used TIDE failure analysis to direct dataset fixes and threshold tuning.
- Defined a repeatable baseline-to-FP16 validation workflow with acceptance thresholds for latency, memory and stability, helping prevent model regressions before field release.
- Applied OpenCV and scikit-image preprocessing and artefact detection to identify mud and sand contamination on camera feeds, reducing false positives by 13%.
- Led on-site validation with rig operators, diagnosed failures under lighting, motion and environmental stress, and converted field observations into training and deployment fixes.
- Hardened Dockerised video services with Kafka messaging, health checks, safe restarts, tests and structured observability, cutting live-pipeline crashes by 67% and enabling remote recovery.

LivNSense Technologies | Lead Software Engineer,

05/2020 - 12/2022 | BENGALURU, INDIA

AI

- Led four engineers building a two-stage worker detector and helmet-colour classifier for construction safety in Texas, cutting false positives by 33% within edge latency and power budgets.
- Owned production CV pipelines from ingestion and labelling through training, optimisation, deployment, monitoring and field validation; set experimentation standards and reviewed model performance.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising video streams and running incident detection under constrained edge compute.
- Added ByteTrack multi-object tracking to preserve worker and vehicle identities across frames, reducing duplicate incident alerts from moving-vehicle feeds.
- Built asynchronous C++17 OpenVINO pipelines across Intel Movidius VPUs and Core i7 industrial edge systems, increasing throughput from 5 to 15 FPS by overlapping pre-processing, inference and post-processing.
- Tuned camera and image signal processor settings across RGB and infrared streams for 24/7 operation, reducing false positives and repeated manual recalibration.
- Ran validation under occlusion, lighting variation and motion artefacts across camera feeds, reducing performance variance by 25%; clustered failures with confusion matrices and TIDE to prioritise data improvements.
- Deployed containerised inference microservices on Azure with REST endpoints and queue-based asynchronous processing, owning autoscaling, monitoring and cost-per-inference decisions.
- Automated labelling with existing detection models and DVC dataset versioning, cutting manual labelling effort by 80%; added synthetic-data tooling that reduced further manual labelling time by 60%.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline for YOLO detection and NvDCF tracking across three concurrent video feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a TensorRT FP16 engine supporting variable batch sizes on 6 GB VRAM Turing hardware, validated FP16 against FP32 accuracy and profiled p99 latency to sustain 29.7 FPS per stream live.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving anchor-free post-processing into the inference engine for custom-layer deployment.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, identity switches, fragmentation, FPS and VRAM usage to make tracker selection evidence-based.

- Developed a repeatable evaluation harness for YOLOv8, YOLOv10, YOLOv26 and RT-DETR, comparing accuracy, latency and model size for edge deployment.
- Extended evaluation to segmentation and depth models, including EfficientViT, MobileSAM and Depth Anything v2, under the same compute constraints.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.
- Exported models to ONNX and tested OpenVINO paths, comparing mAP, inference time and model size to support deployment decisions.

EDUCATION

University of Bristol | MSc Robotics,
Merit

09/2023 - 09/2024 | BRISTOL, UK

Dissertation: Radiological SLAM with LiDAR Odometry

- Developed and evaluated a real-time C++17 and ROS2 perception system on a Clearpath Husky, combining Velodyne LiDAR, RealSense depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer on matched rosbag2 sequences under one evaluation protocol to compare mapping and localisation behaviour.
- Configured Ceres, g2o and GTSAM solver backends and tuned ICP parameters across environments to evaluate mapping behaviour under controlled changes.
- Calibrated sensor extrinsics and resolved timestamp and transform faults with tf2 across a PCL point-cloud pipeline running on NVIDIA Jetson AGX Xavier.

Visvesvaraya National Institute of Technology | BTech Electrical and Electronics
Engineering

07/2016 - 05/2020 | NAGPUR, INDIA

- Studied signals, systems, embedded hardware and control, providing a foundation for real-time AI, sensing and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash, SQL
Computer Vision	object detection, multi-object tracking, image classification, optical flow, semantic and instance segmentation, incident detection, data augmentation, OpenCV, PyTorch, TensorFlow, Keras
Video Analytics	NVIDIA DeepStream 6.x and 9.0, GStreamer, FFmpeg, RTSP, multi-stream processing, multi-camera synchronisation, video metadata extraction, ByteTrack, NvDCF, NvSORT
Models	YOLOv5, YOLOv8, YOLOv10, YOLOv26, RT-DETR, EfficientDet, Mask R-CNN, DeepLabv3+, U-Net, SegFormer
Validation	model benchmarking, dataset validation, COCO annotation format, TIDE failure analysis, confusion matrices, ablation studies, hypothesis testing, accuracy and latency regression testing
Edge Inference	TensorRT, OpenVINO, ONNX, FP16 and INT8 quantisation, structured pruning, variable batch sizes, custom TensorRT plugins, Intel Movidius VPU, NVIDIA Jetson AGX Xavier
MLOps and Data	MLflow, DVC, Apache Airflow, Apache Kafka, CVAT, Optuna, Weights & Biases, automated labelling, dataset versioning, synthetic data, production monitoring, drift tracking
Production	Docker, CI/CD, Git, CMake, pytest, unittest, gtest, REST APIs, microservices, multiprocessing, structured logging, health checks, observability, Linux, Azure
Leadership	technical leadership, mentoring, experiment standards, code review, stakeholder communication, cross-functional delivery, Agile and Scrum

ADDITIONAL EXPERIENCE

GrowthStage | Technical Assessor, Computer Vision and Software
Engineering

05/2026 - PRESENT | REMOTE

Assess candidate depth across computer vision, perception, SLAM and deployment using practical engineering evidence.

CERTIFICATIONS

Deep Learning Specialisation, deeplearning.ai (07/2019)